

Electrostatics

Questions with Fully Worked Answers · FormulaVerse PYQ Bank

Q1. Two point charges $q_1 = 3 \mu\text{C}$ and $q_2 = -4 \mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. The force on charge q_2 is ____ N. (Take $1/4\pi\epsilon_0 = 9 \times 10^9$; SI units)

JEE Main 2026 (Online) 8th April Evening Shift

✓ **Answer: $(0.004\hat{i} + 0.008\hat{j} + 0.008\hat{k})$ N, magnitude 0.012 N**

Solution:

Vector from q_1 to q_2 : $r = (q_2 - q_1 \text{ position}) = -\hat{i} - 2\hat{j} - 2\hat{k}$, with $|r| = \sqrt{1+4+4} = 3 \text{ m}$.

Force magnitude: $F = k|q_1q_2|/r^2 = (9 \times 10^9)(3 \times 10^{-6})(4 \times 10^{-6})/9 = 0.012 \text{ N}$.

Since the charges are unlike, the force on q_2 is attractive — directed back towards q_1 , i.e. along $+\hat{i}+2\hat{j}+2\hat{k}$ (opposite to r).

$F = (0.004\hat{i} + 0.008\hat{j} + 0.008\hat{k}) \text{ N}$, of magnitude 0.012 N (12 mN).

Q2. The electric potential as a function of x, y is given by $V = 5(x^2 - y^2) \text{ V}$. The electric field at a point $(2, 3) \text{ m}$ is ____ V/m.

JEE Main 2026 (Online) 6th April Evening Shift

✓ **Answer: $E = (-20\hat{i} + 30\hat{j}) \text{ V/m}$, magnitude $\approx 36.06 \text{ V/m}$**

Solution:

$E = -\nabla V$, so $E_x = -\partial V/\partial x = -10x$ and $E_y = -\partial V/\partial y = 10y$.

At $(2, 3)$: $E_x = -20 \text{ V/m}$, $E_y = 30 \text{ V/m}$.

$|E| = \sqrt{(20^2+30^2)} = \sqrt{1300} = 10\sqrt{13} \approx 36.06 \text{ V/m}$.

Q3. A capacitor of capacitance 600 pF is charged by a 200 V source. It is then disconnected and connected to another uncharged identical 600 pF capacitor. The electrostatic energy lost in the process is ____ μJ .

JEE Main (Physics, Electrostatic Potential & Capacitance)

✓ **Answer: 6**

Solution:

Initial charge: $Q = CV = 600 \times 10^{-12} \times 200 = 1.2 \times 10^{-7} \text{ C}$.

Initial energy: $U_i = \frac{1}{2}CV^2 = \frac{1}{2}(600\text{pF})(200)^2 = 1.2 \times 10^{-5} \text{ J} = 12 \mu\text{J}$.

When connected to an identical uncharged capacitor, charge is shared equally over total capacitance 1200 pF , giving common voltage $V' = Q/1200\text{pF} = 100 \text{ V}$.

Final energy: $U_f = \frac{1}{2}(1200\text{pF})(100)^2 = 6 \times 10^{-6} \text{ J} = 6 \mu\text{J}$.

Energy lost $= U_i - U_f = 12 - 6 = 6 \mu\text{J}$.
